

Supplemental Material

Table S1. Summary statistics for air pollutions in Seoul, 2007–2013

Pollutants	Mean (SD*)	Quantiles					IQR†
		Min‡	25%	50%	75%	Max§	
PM _{2.5} , µg/m ³	25.03 (14.29)	2.79	15.04	21.92	31.00	121.10	15.96
PM _{2.5-10} , µg/m ³	25.23 (20.99)	2.00	13.52	21.21	30.04	367.30	16.52
CO, ppm	0.57 (0.24)	0.19	0.41	0.51	0.66	1.81	0.25
SO ₂ , ppb#	5.53 (2.30)	2.24	9.92	4.92	6.46	22.52	2.54
NO ₂ , ppb	34.45 (12.92)	6.44	24.52	32.32	42.92	92.35	18.4
O ₃ , ppb	19.93 (10.86)	1.84	11.18	18.75	27.08	71.12	15.9

*SD denotes standard deviation; †IQR, interquartile ranges; ‡min, minimum; §max, maximum; ||ppm, parts-per-million, 10⁻⁶. #ppb, parts-per-billion, 10⁻⁹;

Table S2. Correlation matrix among air pollutants, in Seoul, 2007-2013

	PM _{2.5}	PM _{2.5-10}	CO	SO ₂	NO ₂	O ₃
PM _{2.5}	-	0.58	0.84	0.75	0.69	0.67
PM _{2.5-10}	0.58	-	0.93	0.42	0.45	0.37
CO	0.84	0.93	-	0.62	0.62	0.55
SO ₂	0.75	0.42	0.62	-	0.77	0.84
NO ₂	0.69	0.45	0.62	0.77	-	0.68
O ₃	0.67	0.37	0.55	0.84	0.68	-

Table S3. Incidence of cardiovascular events per 100 person-years in Seoul, 2007–2013

Event	Total	Sex	
		Male	Female
Major adverse cardiovascular events	1,856 (0.21)*	1001 (0.23)	855 (0.19)
All-cause death	1,658 (0.18)	998 (0.23)	660 (0.14)
Cardiovascular death	265 (0.03)	133 (0.03)	132 (0.03)
Acute myocardial infarction	354 (0.04)	208 (0.05)	146 (0.03)
Congestive heart failure	652 (0.07)	319 (0.07)	333 (0.07)
Stroke	934 (0.10)	522 (0.12)	412 (0.09)
Ischemic stroke	688 (0.08)	385 (0.03)	303 (0.07)
Hemorrhagic stroke	292 (0.03)	164 (0.04)	128 (0.03)

*Data were presented as number of events (incidence per 100 person-years).

Figure S1. The yearly trends of the average concentrations of air pollutants 2007-2013

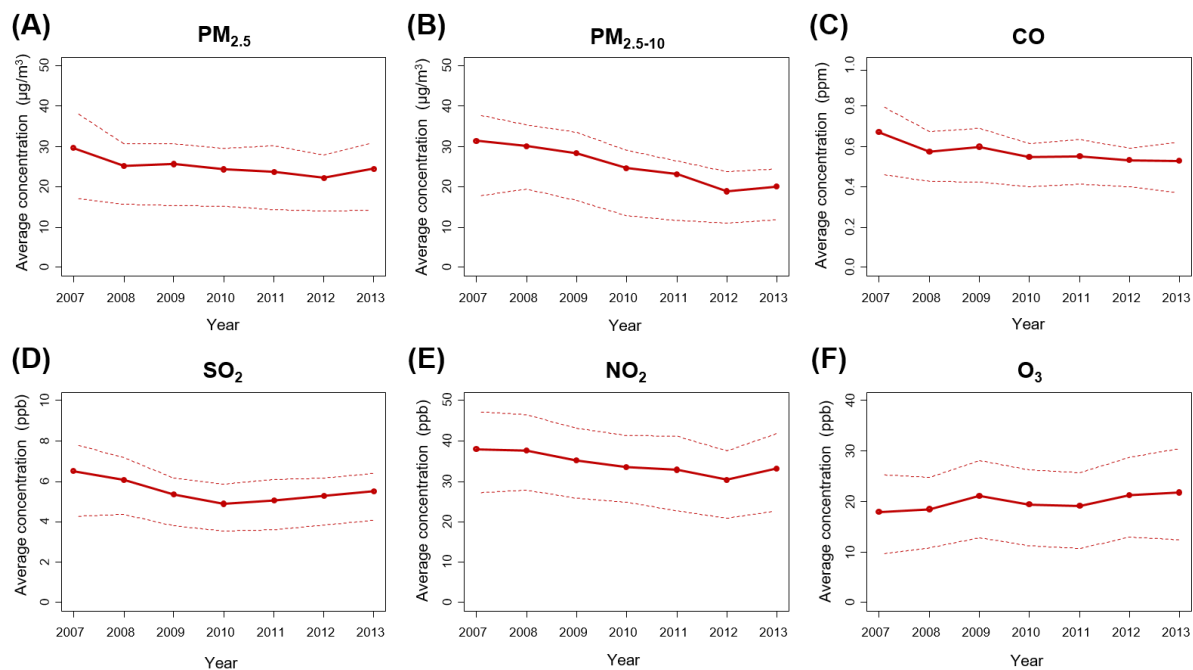


Figure S2. Single, two-, and multi-pollutant models for cardiovascular risk of long-term exposure to PM_{2.5}. The y-axis denotes the hazard ratio (box) with 95% confidence intervals (line).

